Project Team: Rafes Divas (L. Trent, L. Anggara, N. Green)

Field Guide: ICU Mortality Risk Prediction

What This Tool Does?

This tool uses routinely collected ICU data to estimate a patient's risk of death during the admission. It outputs a mortality risk score to help nurses and physicians identify patients who may need closer monitoring, earlier escalation, or a more careful clinical review. It is a support tool only and does not replace clinical judgment.

How It Works

The tool was developed using ICU patient records containing age, comorbidity burden, vital signs, oxygenation, sepsis status, and need for support such as ventilation or vasopressors. It identifies patterns in these data that were associated with death during admission and uses them to generate a new patient's risk score.

Two model types were tested: one simpler and easier to explain, and one more flexible. Both estimate risk from overall data patterns rather than from a single variable. The tool works best when data are complete, current, and similar to the ICU population used in development.

Performance Summary

Table 1. Performance Summary

Metric	Value	Clinical meaning
AUROC	0.614 (Logistic Regression); 0.617 (XGBoost)	Modest ability to distinguish patients who died from those who survived
PR AUC	0.318 (Logistic Regression); 0.304 (XGBoost)	Limited precision for identifying high-risk patients
Precision	0.280 (Logistic Regression); 0.270 (XGBoost)	About 27–28% of patients flagged high risk actually died
Recall	0.737 (Logistic Regression); 0.773 (XGBoost)	The models identified about 74–77% of patients who died
Brier Score	0.171 for both models	Similar overall accuracy of predicted risk

These results were measured on an internal test set from 15,000 ICU patients, with 22.8% hospital mortality. Two models were tested: Logistic Regression and XGBoost. Overall performance was modest, and neither model should be used alone for clinical decisions. XGBoost found slightly more patients who died, while Logistic Regression had slightly better precision. The AUROC confidence intervals overlap, so the difference appears small.

Important: Performance has only been evaluated on the study dataset. Comparison with clinician performance and subgroup analysis were not performed.

Interpreting Output

Table 2. Risk score interpretation

Risk score	Interpretation	Clinical response
0–30%	Lower predicted risk	Do not assume the patient is safe. Compare with the bedside picture and recent trend.
30–70%	Intermediate predicted risk	Use caution. Clinical judgment is especially important.
70–100%	Higher predicted risk	Correlate with the clinical picture. Consider closer review and possible escalation if supported clinically.

Before acting on the score, ask:

- Does the result match the patient's current condition?
- Are the input data complete and believable?
- Is the patient worsening despite a low score?
- Is the patient stable despite a high score?

When to Use / When Not to Use?

Use when:

- Routine ICU clinical data are available.
- The team wants added risk awareness during admission review, rounds, handoff, or escalation discussions.
- The score is being interpreted together with bedside assessment, labs, imaging, and overall clinical trajectory.

Do not use when:

- It would be the only basis for a treatment decision.
- Key data are missing, outdated, or clearly incorrect.
- The patient is outside the population the tool was developed on, unless further validation is available.
- The score would be used alone for limitation-of-care, goals-of-care, or end-of-life decisions.

Limitations and Failure Modes

This tool was tested on one internal ICU dataset only, so performance in other hospitals or patient populations is unknown. It may be less reliable when the input data are missing, delayed, inaccurate, or not reflective of the patient's current condition. It may also perform less well in clinical situations that differ from the study data.

The most likely failure pattern is a false positive, where the patient is flagged as high risk but survives. This may lead to unnecessary concern or escalation. It can also produce false negatives, where a patient who later dies is not flagged as high risk, creating false reassurance. Because overall performance was modest, both types of error should be expected.

Human Override Rules

Table 3. Human Override Rules

Situation	What to do
High-risk score, but low clinical concern	Recheck the data and reassess the patient. Do not escalate based on the score alone.
Low-risk score, but high clinical concern	Follow the bedside picture. Do not let a low score override concern about deterioration.
Missing or unreliable data	Do not rely on the score until the data are verified.
Patient outside validated population	Use extra caution and treat the result as supplemental only.
High-stakes decisions	Never use the score alone for major treatment or goals-of-care decisions.
Uncertain case	Escalate to senior review, repeat assessment, and continue monitoring.

Integration with Clinical Workflow

This tool fits best into routine ICU care as an added risk-awareness step. It can be reviewed during admission, rounds, handoff, or changes in patient status. The score should always be interpreted together with bedside assessment, vital sign trends, labs, organ support needs, and overall trajectory. Its main role is to prompt closer review, not to make decisions on its own.

Training Requirements

Users should receive basic training before using the tool. They should understand what the score means, what data go into it, when it should and should not be used, and when it must be overridden. Training should include example cases, common failure scenarios, and clear instruction that the score must never replace clinical judgment in high-stakes decisions.

Reporting Issues

Report technical problems, unexpected outputs, or suspected unsafe behavior through the project's designated reporting pathway. Include the patient context, input concerns, output shown, and why the result appeared inconsistent with the clinical picture.